

B6 Condensed-Matter Physics — Model Solutions, 2018

Question 1 — Crystal structure of tungsten

Problem. Identify lattice types of α - and β -tungsten; from the lowest-angle powder XRD ring of α -W (0.1542 nm X-rays, $2\theta = 40.26^\circ$) deduce a and the density; for β -W find the first two non-vanishing rings, their intensity ratio, and the lowest-angle additional weak ring expected.

(a) Definition and lattice type of α -W

A *lattice* is the infinite set of points

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3, \quad n_i \in \mathbb{Z},$$

generated by three primitive vectors $\mathbf{a}_i$; every lattice point has identical environment.

For α -W, the two atoms at $[0, 0, 0]$ and $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ are chemically identical and have identical local environments, so each is a lattice point. The conventional cubic cell with the centring vector $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ is

body-centred cubic (BCC).

(b) Lattice constant and density

Selection rule (BCC): reflections allowed for $h + k + l$ even; lowest-sin θ ring is (110).

Bragg's law:

$$\lambda = 2d_{110} \sin \theta.$$

Solve for d_{110} :

$$d_{110} = \frac{\lambda}{2 \sin \theta} = \frac{0.1542 \text{ nm}}{2 \sin(20.13^\circ)}.$$

Evaluate denominator: $2 \sin(20.13^\circ) = 0.6884$:

$$d_{110} = 0.2240 \text{ nm}.$$

Cubic spacing $d_{hkl} = a / \sqrt{h^2 + k^2 + l^2}$:

$$a = d_{110} \sqrt{2}.$$

Numerical:

$$a = 0.3168 \text{ nm}.$$

BCC conventional cell contains 2 atoms. Mass density:

$$\rho = \frac{2M}{N_A a^3}.$$

Insert numbers:

$$\rho = \frac{2 \times 0.1838 \text{ kg mol}^{-1}}{(6.022 \times 10^{23} \text{ mol}^{-1}) (3.168 \times 10^{-10} \text{ m})^3}.$$

Evaluate a^3 : $a^3 = 3.179 \times 10^{-29} \text{ m}^3$:

$$\rho = \frac{0.3676 \text{ kg mol}^{-1}}{1.914 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}}.$$

Final:

$$\rho \approx 1.92 \times 10^4 \text{ kg m}^{-3} = 19.2 \text{ g cm}^{-3}.$$

(c) Lattice type of β -W

The two corner/centre atoms $\{[0, 0, 0], [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]\}$ have a different local environment from the six face-pair atoms $[\frac{1}{4}, \frac{1}{2}, 0]$ etc., so the eight sites are *not* all equivalent: they cannot all be lattice points.

Test BCC centring $\mathbf{T} = [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$:

$$[\frac{1}{4}, \frac{1}{2}, 0] + [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}] = [\frac{3}{4}, 1, \frac{1}{2}] \equiv [\frac{3}{4}, 0, \frac{1}{2}],$$

which is not in the basis: BCC ruled out. FCC and primitive face centring similarly fail. Hence

simple (primitive) cubic, with an 8-atom basis (A15 structure).

(d) First two rings of β -W and their intensity ratio

All eight atoms have the same form factor f . Structure factor:

$$S_{hkl} = f \sum_{j=1}^8 e^{2\pi i(hx_j + ky_j + lz_j)}.$$

Group corner/centre pair $\times$ face-pair triplet:

$$S_{hkl} = f[1 + (-1)^{h+k+l}] \left\{ 1 + 2 \cos \frac{\pi h}{2} \cos \pi k + 2 \cos \frac{\pi l}{2} \cos \pi h + 2 \cos \frac{\pi k}{2} \cos \pi l \right\}.$$

First bracket vanishes for $h + k + l$ odd: (100), (110), (111) absent.

Tabulate $|S/f|^2$ and multiplicity m :

(hkl)	$h^2 + k^2 + l^2$	$ S/f ^2$	m
(200)	4	16	6
(210)	5	16	24
(211)	6	16	24
(220)	8	0	—

First two non-vanishing rings:

(200) and (210).

Both have $|S|^2 = 16f^2$, so the ratio is set by multiplicity (Lorentz-polarisation and Debye-Waller corrections are subleading):

$$\frac{I_{210}}{I_{200}} \approx \frac{m_{210}}{m_{200}} = \frac{24}{6}.$$

$$\frac{I_{210}}{I_{200}} \approx 4.$$

(e) Additional weak ring at lower angle

The cancellations producing $|S| = 0$ for $h+k+l$ odd assumed all 8 sites have identical scattering. The two Wyckoff sites are chemically inequivalent: corner/centre atoms vibrate differently from face-pair atoms, giving distinct anisotropic Debye–Waller factors and slightly different effective form factors, $f_1 \neq f_2$.

With $f_1 \neq f_2$ the structure factor becomes $S = (f_1 - f_2) \times [\text{previously-vanishing prefactor}]$, so previously forbidden rings reappear with intensity $\propto (f_1 - f_2)^2 \ll (f_1 + f_2)^2$.

Lowest forbidden ring (smallest $h^2 + k^2 + l^2$): (100). However (100) also vanishes from the second bracket on the corner/centre sublattice; the lowest ring with non-zero contrast between the two sublattices is

$$(110),$$

seen weakly because $|f_1 - f_2| \ll f$.

Question 2 — Tight binding

Problem. Define μ and E_F . For a 1D tight-binding chain with hopping $-t$, derive the dispersion, sketch it and the DOS. For monovalent atoms find E_F and show $\mu(T) = E_F$. Repeat the Fermi-surface and $\mu(T)$ analysis on a 2D square lattice, then on a squeezed rectangular lattice.

(a) Definitions

Chemical potential: $\mu = (\partial U / \partial N)_{S,V}$, the energy cost of adding one particle at fixed entropy and volume; equivalently, the energy at which the Fermi–Dirac function equals $\frac{1}{2}$.

Fermi energy: $E_F = \mu(T = 0)$, the energy of the highest occupied single-particle state in the ground state of a non-interacting fermion system.

(b) 1D dispersion, BZ diagram, DOS

Expand a single-particle eigenstate $|\psi\rangle = \sum_n \psi_n |n\rangle$. Apply H :

$$H|\psi\rangle = \sum_n \psi_n (\epsilon_0 |n\rangle - t |n+1\rangle - t |n-1\rangle).$$

Project onto $\langle m|$:

$$E \psi_m = \epsilon_0 \psi_m - t \psi_{m-1} - t \psi_{m+1}.$$

Bloch ansatz $\psi_n = e^{ikna}$:

$$E = \epsilon_0 - t e^{-ika} - t e^{ika}.$$

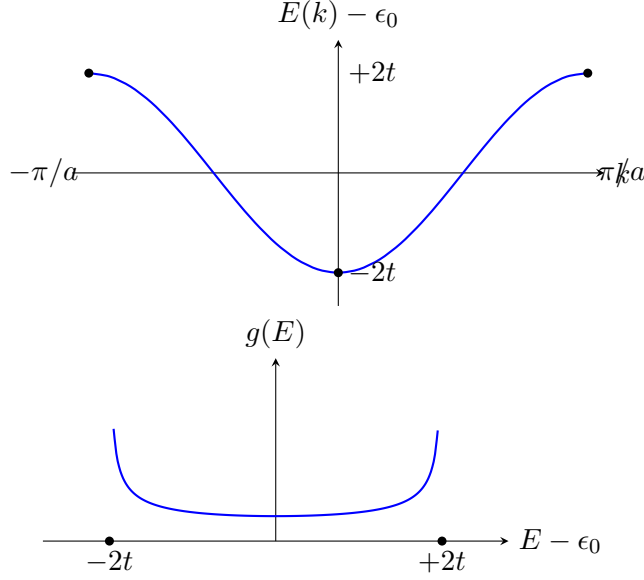
Combine exponentials:

$$E(k) = \epsilon_0 - 2t \cos(ka), \quad k \in (-\pi/a, \pi/a].$$

DOS in 1D follows from $g(E) dE = (L/\pi) dk$ (factor 2 for spin already absorbed by per-atom counting):

$$g(E) = \frac{L}{\pi} \frac{1}{|dE/dk|} = \frac{L}{\pi a} \frac{1}{\sqrt{(2t)^2 - (E - \epsilon_0)^2}}.$$

Van Hove singularities at $E = \epsilon_0 \pm 2t$.



(c) Half-filling: E_F and $\mu(T)$

Bandwidth $4t$, N atoms, $2N$ spin states, N electrons (monovalent): half-filled. Symmetry $E(k + \pi/a) - \epsilon_0 = -[E(k) - \epsilon_0]$ gives $g(\epsilon_0 + \delta) = g(\epsilon_0 - \delta)$.

At $T = 0$, fill the lower half of the band:

$$E_F = \epsilon_0.$$

Particle number constraint at any T :

$$N_e = \int g(E) n_F(E, \mu, T) dE = \frac{1}{2} \int g(E) dE.$$

Apply hint $n_F(E, \mu) = 1 - n_F(2\mu - E, \mu)$:

$$\int g(E) n_F(E, \mu) dE = \int g(E) dE - \int g(E) n_F(2\mu - E, \mu) dE.$$

Substitute $E \rightarrow 2\mu - E$ in the second integral, with $g(2\mu - E)$ appearing:

$$\frac{1}{2} \int g(E) dE = \int g(E) dE - \int g(2\mu - E) n_F(E, \mu) dE.$$

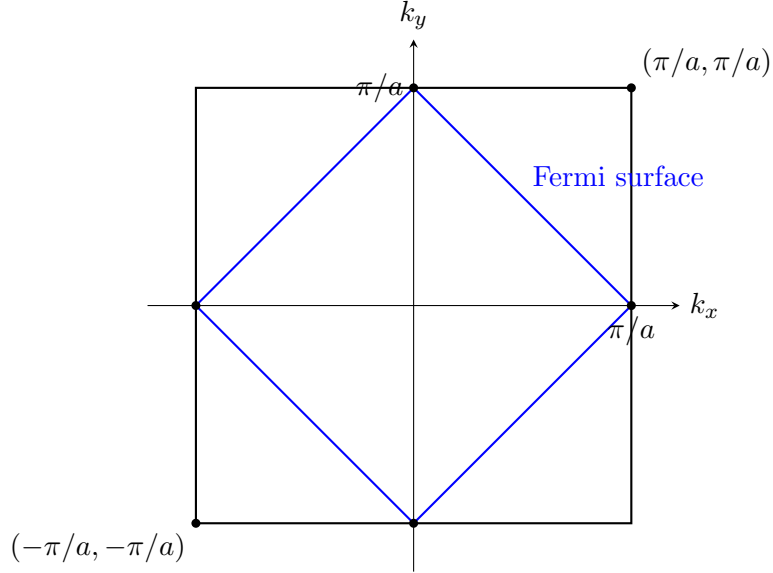
Demand consistency with $g(2\epsilon_0 - E) = g(E)$: the equation collapses to an identity iff $\mu = \epsilon_0$:

$$\mu(T) = \epsilon_0 = E_F \quad \text{for all } T.$$

Particle-hole symmetry pins μ to the band centre.

(d) 2D square lattice, monovalent

Brillouin zone: square $-\pi/a < k_x, k_y \leq \pi/a$. Half-filling sets $\cos(k_x a) + \cos(k_y a) = 0$: a square rotated by 45° , vertices $(\pm\pi/a, 0), (0, \pm\pi/a)$.



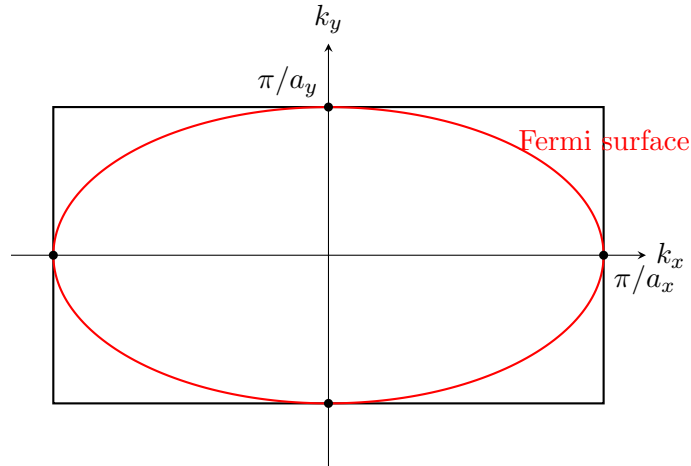
Nesting $E(\mathbf{k} + \mathbf{Q}) + E(\mathbf{k}) = 2\epsilon_0$ with $\mathbf{Q} = (\pi/a, \pi/a)$ gives the same particle-hole symmetry as in (c):

$$\mu(T) = \epsilon_0 \quad \text{for all } T.$$

(e) Squeezed rectangular lattice, $a_x < a_y$

Brillouin zone is rectangular and longer along k_x (since $\pi/a_x > \pi/a_y$). Hopping anisotropy: shorter bond a_x has larger overlap, so $t_x > t_y$; dispersion $E = \epsilon_0 - 2t_x \cos(k_x a_x) - 2t_y \cos(k_y a_y)$.

Half-filling locus passes through fixed points $(\pm\pi/a_x, 0)$ and $(0, \pm\pi/a_y)$ (band-centre by symmetry); curvature distorts the diamond to a closed convex curve, bulging outward in k_y (slow dispersion, narrow band) and pinched inward along k_x .



Nesting at $\mathbf{Q} = (\pi/a_x, \pi/a_y)$ is preserved, hence again $\mu(T) = \epsilon_0$.

Question 3 — Phonons in lead

Problem. Define phonon and Brillouin zone. From the Morse potential ($\xi = 0.084 \text{ nm}$, $A = 1600 \text{ eV}$, $B = 40 \text{ eV}$) compute the equilibrium bond length a , binding energy and harmonic spring

constant κ . Compute the 1D sound velocity (Pb, $M = 207 \text{ g mol}^{-1}$). Estimate the thermal-expansion coefficient using cubic anharmonicity, and identify the validity ranges in T .

(a) Definitions

Phonon: a quantised normal mode of lattice vibration carrying energy $\hbar\omega$ and crystal momentum $\hbar\mathbf{k}$, obeying Bose–Einstein statistics.

Brillouin zone: the Wigner–Seitz primitive cell of the reciprocal lattice; equivalently, the set of $\mathbf{k}$ closer to the origin than to any other reciprocal lattice point.

(b) Morse: $a, V_{\min}, \kappa$

Set $u \equiv e^{-r/\xi}$:

$$V = Au^2 - Bu.$$

Differentiate w.r.t. r via $du/dr = -u/\xi$:

$$V'(r) = -\frac{u}{\xi}(2Au - B).$$

Equilibrium $V'(a) = 0$:

$$u_a = \frac{B}{2A}.$$

Invert $u_a = e^{-a/\xi}$:

$$a = \xi \ln \frac{2A}{B}.$$

Numerical: $2A/B = 80$, $\ln 80 = 4.382$:

$$a = 0.084 \text{ nm} \times 4.382 = 0.368 \text{ nm}.$$

Binding energy: insert $u_a = B/(2A)$ into V :

$$V(a) = A \left(\frac{B}{2A} \right)^2 - B \cdot \frac{B}{2A} = \frac{B^2}{4A} - \frac{B^2}{2A}.$$

Simplify:

$$V(a) = -\frac{B^2}{4A} = -\frac{(40)^2}{4 \cdot 1600} \text{ eV} = -0.250 \text{ eV}.$$

Second derivative. Differentiate $V'(r) = -(u/\xi)(2Au - B)$ once more:

$$V''(r) = \frac{1}{\xi^2}(4Au^2 - Bu).$$

Evaluate at $u = u_a$:

$$\kappa \equiv V''(a) = \frac{1}{\xi^2} \left(4A \cdot \frac{B^2}{4A^2} - B \cdot \frac{B}{2A} \right) = \frac{B^2}{2A\xi^2}.$$

Numerical (in eV per nm²):

$$\kappa = \frac{40^2}{2 \cdot 1600 \cdot (0.084)^2} \text{ eV nm}^{-2} = 70.9 \text{ eV nm}^{-2}.$$

Convert to SI: $1 \text{ eV nm}^{-2} = 1.602 \times 10^{-19} \text{ J}/10^{-18} \text{ m}^2 = 0.1602 \text{ N m}^{-1}$:

$$\kappa \approx 11.4 \text{ N m}^{-1}.$$

(c) Sound velocity

Monatomic chain dispersion:

$$\omega(k) = 2\sqrt{\kappa/m} |\sin(ka/2)|.$$

Long-wavelength group velocity $v_s = d\omega/dk|_{k \rightarrow 0}$:

$$v_s = a\sqrt{\kappa/m}.$$

Atomic mass:

$$m = \frac{M}{N_A} = \frac{0.207 \text{ kg mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}} = 3.44 \times 10^{-25} \text{ kg}.$$

Compute κ/m :

$$\frac{\kappa}{m} = \frac{11.4}{3.44 \times 10^{-25}} \text{ s}^{-2} = 3.31 \times 10^{25} \text{ s}^{-2}.$$

Square root:

$$\sqrt{\kappa/m} = 5.76 \times 10^{12} \text{ s}^{-1}.$$

Multiply by a :

$$v_s = (0.368 \times 10^{-9} \text{ m}) \times (5.76 \times 10^{12} \text{ s}^{-1}) \approx 2.1 \times 10^3 \text{ m s}^{-1}.$$

Compare measured Pb longitudinal speed $\sim 2000 \text{ m s}^{-1}$: agreement.

(d) Thermal expansion from cubic anharmonicity

Expand V to cubic order around $r = a$, with $x \equiv r - a$:

$$V(r) \approx V(a) + \frac{1}{2}\kappa x^2 + \frac{1}{6}\kappa_3 x^3, \quad \kappa_3 \equiv V'''(a).$$

Differentiate V'' once more:

$$V'''(r) = -\frac{1}{\xi^3}(8Au^2 - Bu).$$

Insert $u_a = B/(2A)$:

$$\kappa_3 = -\frac{1}{\xi^3} \left(\frac{2B^2}{A} - \frac{B^2}{2A} \right) = -\frac{3B^2}{2A\xi^3} < 0.$$

Treat cubic term as perturbation; harmonic averages $\langle x \rangle_0 = 0$, $\langle x^2 \rangle_0 = k_B T / \kappa$, $\langle x^3 \rangle_0 = 0$, $\langle x^4 \rangle_0 = 3(k_B T / \kappa)^2$.

First-order in κ_3 :

$$\langle x \rangle \approx -\frac{\kappa_3}{6k_B T} \langle x^4 \rangle_0.$$

Substitute $\langle x^4 \rangle_0$:

$$\langle x \rangle = -\frac{\kappa_3}{6k_B T} \cdot 3 \left(\frac{k_B T}{\kappa} \right)^2 = -\frac{\kappa_3 k_B T}{2\kappa^2}.$$

Differentiate w.r.t. T and divide by a :

$$\alpha = \frac{1}{a} \frac{d\langle r \rangle}{dT} \sim \frac{|\kappa_3| k_B}{\kappa^2 a}.$$

Numerical sanity check: $|\kappa_3| = 3B^2/(2A\xi^3)$. Convert to SI: $\kappa_3 \sim 4 \times 10^{-9} \text{ J m}^{-3}$ (i.e. $\sim 4 \times 10^9 \text{ N m}^{-2}$ in spring units). With $\kappa = 11.4 \text{ N m}^{-1}$, $a = 0.368 \text{ nm}$, $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$:

$$\alpha \sim \frac{(4 \times 10^9 \text{ N m}^{-2})(1.38 \times 10^{-23} \text{ J K}^{-1})}{(11.4 \text{ N m}^{-1})^2 (3.68 \times 10^{-10} \text{ m})} \sim 10^{-4} \text{ K}^{-1}.$$

Order-of-magnitude correct (measured $\alpha_{\text{Pb}} \approx 2.9 \times 10^{-5} \text{ K}^{-1}$); positive sign because $\kappa_3 < 0$ widens the well asymmetrically outwards.

(e) Validity ranges

(i) Cubic-only expansion. Require harmonic $\gg$ cubic energy scale, $\langle \frac{1}{2}\kappa x^2 \rangle \gg \langle \frac{1}{6}|\kappa_3|x^3 \rangle$ at thermal amplitude $|x| \sim \sqrt{k_B T / \kappa}$:

$$\kappa \gg \frac{|\kappa_3|}{3} \sqrt{k_B T / \kappa}.$$

Square and rearrange:

$$k_B T \ll \frac{9\kappa^3}{\kappa_3^2}.$$

Numerically this gives $T \ll 10^3$ K; the physical upper bound is set by the well depth $|V(a)|/k_B \approx 2900$ K (or in practice the lead melting point $T_m \approx 600$ K):

$$T \lesssim T_m \approx 600 \text{ K}.$$

(ii) Quantum mechanics important. Compare $\hbar\omega$ to $k_B T$ with $\omega = \sqrt{\kappa/m} = 5.76 \times 10^{12} \text{ s}^{-1}$:

$$T_Q \sim \frac{\hbar\omega}{k_B} = \frac{(1.055 \times 10^{-34} \text{ J s})(5.76 \times 10^{12} \text{ s}^{-1})}{1.381 \times 10^{-23} \text{ J K}^{-1}}.$$

Evaluate:

$$T_Q \approx 40 \text{ K}.$$

For $T \lesssim T_Q$ zero-point motion dominates ($\langle x^2 \rangle \rightarrow \hbar/(2m\omega)$), heat capacity drops below k_B/mode (Debye/Einstein freeze-out), and $\alpha \propto T^3$ at low T (Grüneisen).

Classical cubic-only treatment is therefore valid in $T_Q \lesssim T \lesssim T_m$, i.e. roughly $40 \text{ K} \lesssim T \lesssim 600 \text{ K}$.

Question 4 — Magnetism of GdN

Problem. Define paramagnet, diamagnet, ferromagnet (with examples). Find S of Gd^{3+} ($4f^7$) by Hund. Treat Gd^{3+} as Ising along $\hat{z}$ in FCC GdN ($a = 0.50 \text{ nm}$): compute Curie constant C for $\chi = C/T$, modify χ for $T > T_c = 70 \text{ K}$, give $M(T = 0)$, and sketch $M(T)$ for $0 \leq T < 2T_c$ at $H_z = 0$ and $H_z > 0$.

(a) Definitions

Paramagnet: $\chi > 0$, no spontaneous order; high- T Curie law $\chi \propto 1/T$. Example: MnSO_4 (or any rare-earth salt above its ordering T).

Diamagnet: $\chi < 0$, repelled from field, Larmor response of closed shells. Example: Cu, Bi, water.

Ferromagnet: spontaneous magnetisation $M \neq 0$ below Curie T_c . Example: Fe, Ni, Co.

(b) Spin of Gd^{3+}

Configuration $4f^7$: half-filled shell. Hund's first rule maximises S , all 7 spins parallel:

$$S = 7/2.$$

(Hund 2: $L = 0$; Hund 3: $J = S = 7/2$, $g_J = 2$.)

(c) Curie constant

Ising along $\hat{z}$ with $S = 7/2$ and $g = 2$: each ion is a two-level system with moment

$$m = g S \mu_B = 2 \cdot \frac{7}{2} \mu_B = 7\mu_B.$$

Energies $\mp mB$ in field $B = \mu_0 H_z$. Two-level partition function:

$$Z = 2 \cosh(\beta m B), \quad \langle m_z \rangle = m \tanh(\beta m B).$$

Linearise for $\beta m B \ll 1$:

$$\langle m_z \rangle \approx \frac{m^2 B}{k_B T} = \frac{m^2 \mu_0 H_z}{k_B T}.$$

Multiply by number density:

$$M_z = n \langle m_z \rangle = \frac{n m^2 \mu_0}{k_B T} H_z.$$

Read off:

$$\chi = \frac{n m^2 \mu_0}{k_B T} = \frac{C}{T}, \quad C = \frac{n m^2 \mu_0}{k_B}.$$

Number density: GdN is FCC with one formula unit per primitive cell, equivalently 4 Gd^{3+} per conventional cell of side a :

$$n = \frac{4}{a^3} = \frac{4}{(0.50 \times 10^{-9} \text{ m})^3} = 3.20 \times 10^{28} \text{ m}^{-3}.$$

Insert numbers (with $\mu_B = 9.274 \times 10^{-24} \text{ J T}^{-1}$, $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$):

$$C = \frac{(3.20 \times 10^{28} \text{ m}^{-3})(7 \cdot 9.274 \times 10^{-24} \text{ J T}^{-1})^2 (4\pi \times 10^{-7} \text{ T m A}^{-1})}{1.381 \times 10^{-23} \text{ J K}^{-1}}.$$

Evaluate $(7\mu_B)^2 = (6.49 \times 10^{-23} \text{ J T}^{-1})^2 = 4.21 \times 10^{-45} \text{ J}^2 \text{ T}^{-2}$:

$$C = \frac{(3.20 \times 10^{28})(4.21 \times 10^{-45})(1.257 \times 10^{-6})}{1.381 \times 10^{-23}} \text{ K}.$$

Multiply numerator: 1.69×10^{-22} :

$$\boxed{C \approx 12.3 \text{ K}.$$

(d) Curie–Weiss modification and $M(0)$

Mean-field treatment of exchange: replace T by $T - T_c$:

$$\chi(T) = \frac{C}{T - T_c} \quad \text{for } T > T_c.$$

Denominator vanishes at T_c , signalling spontaneous order.

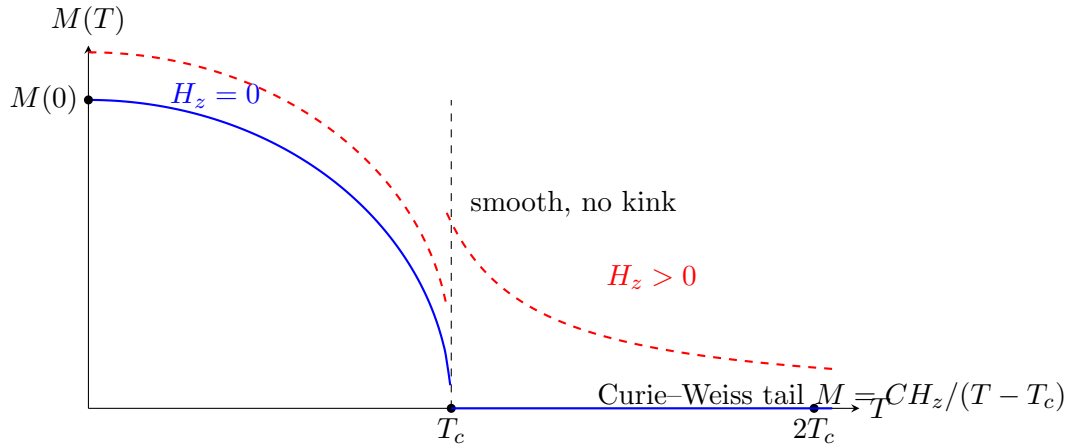
At $T = 0$ Ising spins fully align: $M(0) = nm$. Insert:

$$M(0) = (3.20 \times 10^{28} \text{ m}^{-3}) \times 7 \times (9.274 \times 10^{-24} \text{ J T}^{-1}).$$

Evaluate:

$$\boxed{M(0) \approx 2.1 \times 10^6 \text{ A m}^{-1}.$$

(e) $M(T)$ sketch



Differences:

- $H_z = 0$: spontaneous M only for $T < T_c$; $M \propto (T_c - T)^{1/2}$ near T_c in mean-field; identically zero for $T \geq T_c$.
- $H_z > 0$: field selects the $+\hat{z}$ branch, giving $M > 0$ everywhere; sharp kink at T_c smeared into a crossover; above T_c a Curie-Weiss tail $M = CH_z / (T - T_c) \rightarrow 0$ only as $T \rightarrow \infty$.